

Knowledge Graph e CPS in Sanità

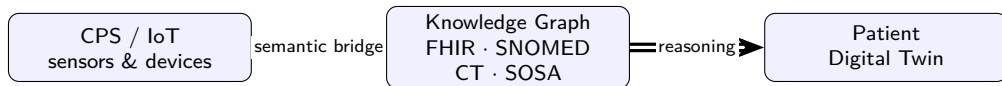
PhD Lecture Module

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The Semantic Bridge Thesis



- Heterogeneous CPS/IoT streams are **syntactically rich but semantically poor**; KGs supply the missing meaning layer.
- A KG makes observations **queryable, reason-able, and clinically comparable** — the substrate on which a Twin is simulated.
- **Lecture thesis:** the *quality of the bridge* (ontologies + pipelines + reasoning), not the sensors, determines clinical trustworthiness.

i Discussion Prompt for PhD Students

If the KG is the bridge, where exactly does it break first in a real ICU — at ingestion, at alignment, or at reasoning? Defend your claim with one failure mode.

Reference. Carbonaro, A., Marfoggia, A., Nardini, F., & Mellone, S. (2023). *CONNECTED: Leveraging digital twins and personal knowledge graphs in healthcare digitalization*. **Frontiers in Digital Health**, 5, Article 1322428. <https://doi.org/10.3389/fdth.2023.1322428>

Roadmap & Time Budget

Section	Focus	Time
1. Foundations & Standards	Ontology stack, architectures, interop limits	10 min
2. CPS-to-KG Integration Pipelines	Sensor→triple architectures, real-time vs batch	15 min
3. Digital Twin & AI Reasoning	Rule-based / neuro-symbolic / RAG, validation	12 min
4. Open Challenges & Frontiers	Scalability, privacy, benchmarks, PhD topics	8 min

- Eight papers (2023–2026) are **re-cut thematically**, not presented one-by-one.
- Watch for a recurring tension: **standards completeness vs. real-time clinical latency**.

Discussion Prompt for PhD Students

Which of the four sections hides the PhD-defining problem? Predict now, then revisit at the end — has your answer changed?

1 · Foundations & Standards

10 minutes — “*What language do the sensors speak?*”

- Goal: establish the **ontology vocabulary** and the **conceptual architecture** before any pipeline.
- Anchored by **Hendawi & Li (2024)** — clinical terminologies; **Chatterjee et al. (2024)** — sensor ontologies; **Carbonaro et al. (2023)** — blueprint.

i Discussion Prompt for PhD Students

Three standards bodies (HL7, W3C, IHTSDO/SNOMED) ship at different speeds. Whose release cadence becomes the bottleneck for a 2026 clinical deployment?

Reference. Hendawi, R., & Li, J. (2024). *Comprehensive personal health knowledge graph for effective management and utilization of personal health data*. In **Proceedings of IEEE AIMHC** (pp. 1–8). IEEE. <https://doi.org/10.1109/aimhc59811.2024.00026>

Healthcare Ontology Foundations: From Codes to Graphs

- **Core contribution:** A *Comprehensive Personal Health KG (PHKG)* unifying EHR, wearables, insurance, and **social determinants of health (SDoH)** under one OWL schema with **SNOMED CT + ICD-10** mapped to nodes.
- Privacy is **schema-native** (role-based access control embedded in the KG), not bolted on; SPARQL exposes **longitudinal/temporal** queries.
- **Ontology used:** OWL · SNOMED CT (clinical) · ICD-10 (administrative).

i Discussion Prompt for PhD Students

SNOMED CT (post-coordinated clinical detail) and ICD-10 (billing granularity) describe the *same patient* at different resolutions. Where does information irreversibly leak in this mapping — and can SDoH (no dominant vocabulary) ever be first-class?

Reference. Hendawi, R., & Li, J. (2024). *Comprehensive personal health knowledge graph for effective management and utilization of personal health data*. In **Proceedings of IEEE AIMHC** (pp. 1–8). IEEE. <https://doi.org/10.1109/aimhc59811.2024.00026>

Sensor Semantics: The SSN/SOSA Bridge

- **Core contribution:** Annotates **MOX2-5 accelerometer** observations as OWL/RDF triples via W3C **SSN/SOSA**; converts **CSV** → **RDF** and compares real vs. synthetic activity data through SPARQL.
- Formally models **sensor** · **observation** · **feature-of-interest** · **result**, yielding device-agnostic, interoperable activity graphs with an explicit **FHIR integration pathway**.
- **Ontology used:** W3C SSN/SOSA (observation structure), candidate FHIR alignment.

Discussion Prompt for PhD Students

SSN/SOSA describes *how* an observation was made but not *what it means clinically*. Where, exactly, does the SOSA-to-FHIR/Observation mapping lose the physiological context the Twin needs?

Reference. Chatterjee, A., Gerdes, M., Prinz, A., Riegler, M., & Martínez, S. (2024). *Semantic representation and comparative analysis of physical activity sensor observations using MOX2-5 sensor in real and synthetic datasets: A proof-of-concept study*. **Scientific Reports**, 14, Article 55183. <https://doi.org/10.1038/s41598-024-55183-6>

Conceptual Blueprint: CONNECTED (PKG + DT)

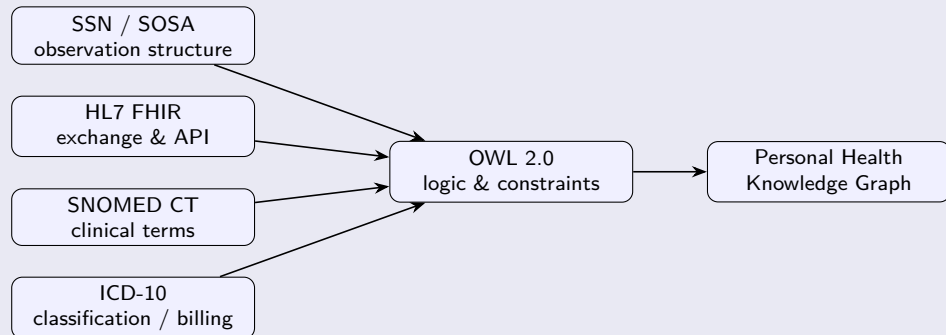
- **Core contribution (seminal 2023):** A systematic review distilled into a **layered architecture** — CPS/IoT acquisition → OWL/FHIR semantic alignment → **Personal Knowledge Graph** (evolving) → **DT simulation**.
- Formalises four requirements rarely co-located: **semantic interoperability, privacy-by-design, real-time data fusion, temporal reasoning**.
- **Ontology used:** OWL · FHIR (alignment layer); DT engine sits *above* the PKG.

i Discussion Prompt for PhD Students

CONNECTED is conceptual with no reference implementation. Which of its four requirements is *most* under-specified today — and would a federated DT-to-PKG link even be auditable under GDPR Art. 22 (automated decisions)?

Reference. Carbonaro, A., Marfoggia, A., Nardini, F., & Mellone, S. (2023). *CONNECTED: Leveraging digital twins and personal knowledge graphs in healthcare digitalization*. **Frontiers in Digital Health**, 5, Article 1322428. <https://doi.org/10.3389/fdgth.2023.1322428>

Ontology Stacking: One Patient, Many Vocabularies



- **FHIR** = exchange spine · **SNOMED CT** = clinical semantics · **ICD-10** = administrative granularity · **SSN/SOSA** = observation provenance · **OWL** = the logic that binds them.
- No single standard is canonical; the **KG is where they are reconciled** — which makes alignment the failure-prone bottleneck.

i Discussion Prompt for PhD Students

If you could declare exactly **one** vocabulary the “spine” of a national PDT platform, which do you

2 · CPS-to-KG Integration Pipelines

15 minutes — “From a voltage on a wrist to a triple in a graph”

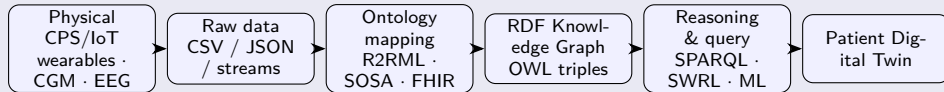
- The **bridge** is itself an engineered artifact: ingestion → mapping → triple materialisation → query.
- Four architectures, one spectrum: **how much happens at the edge vs. centrally, and how often.**

i Discussion Prompt for PhD Students

Before seeing the pipelines: design your own real-time vs. batch cut-over point. What is the maximum latency a hypoglycaemia-alerting pipeline can tolerate before it is clinically useless?

Reference. Pourvahab, M., Monteiro, A., Pais, S., & Pombo, N. (2025). *A unified semantic framework for IoT-healthcare data interoperability: A graph-based machine learning approach using RDF and R2RML*. In **Proceedings of the ACM Symposium on Applied Computing** (Article 3757054). ACM. <https://doi.org/10.1145/3756681.3757054>

The Generic Sensor-to-Triple Pipeline

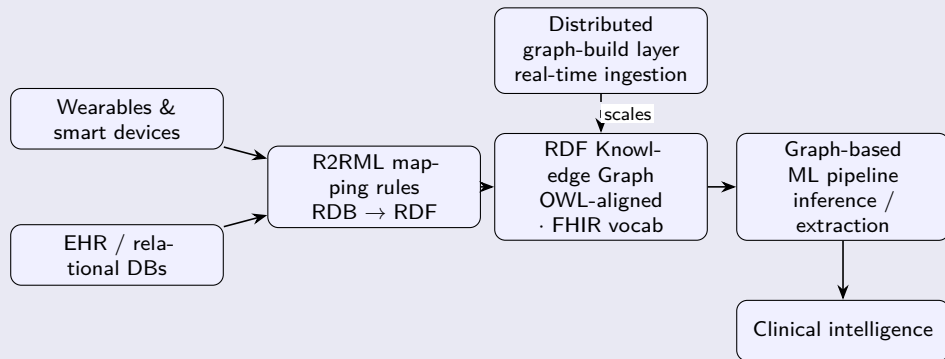


- Every Section-2 paper is an **instantiation** of stages B→D; they differ in *mapping language*, where *materialisation occurs*, and *update frequency*.
- The honest engineering question is **not** “RDF or not?” but “real-time materialisation vs. batch re-build?”

i Discussion Prompt for PhD Students

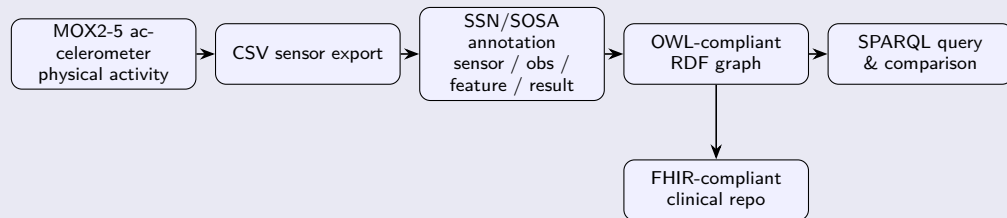
At which stage would you insert a **provenance/quality stamp** (e.g., sensor calibration, trust score)? Argue why doing it earlier or later changes the downstream reasoning.

References. Pourvahab et al. (2025), <https://doi.org/10.1145/3756681.3757054> · Chatterjee et al. (2024), <https://doi.org/10.1038/s41598-024-55183-6>



- **Architecture (Pourvahab et al., 2025):** R2RML declaratively maps relational healthcare data into OWL-aligned triples; a **distributed graph-construction layer** absorbs real-time IoT volume.
- **Transformation steps:** relational row → R2RML template → RDF triple → OWL-classified node → FHIR-compatible vocabulary.
- **Real-time vs. batch trade-off:** *ingestion* is real-time, but the **graph-ML inference** is batch over the constructed KG → low-latency alerts need a streaming bypass.

Wearable CPS-to-KG Construction (SSN/SOSA)

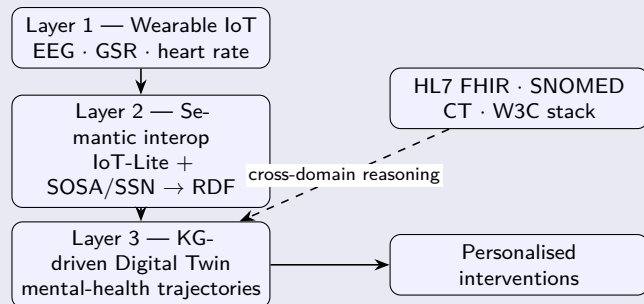


- **Architecture (Chatterjee et al., 2024):** a **concrete CSV→RDF** pipeline — the most explicit “how to triplify a wearable” reference in the set.
- **Transformation steps:** raw accelerometry → activity classification → SOSA `sosa:Observation` with `hasResult` → linked to `FeatureOfInterest` (patient).
- **Real-time vs. batch trade-off:** proof-of-concept is **offline/batch** (file export); no native streaming → gains simplicity & reproducibility, loses latency, so unsuitable as-is for acute monitoring.

i Discussion Prompt for PhD Students

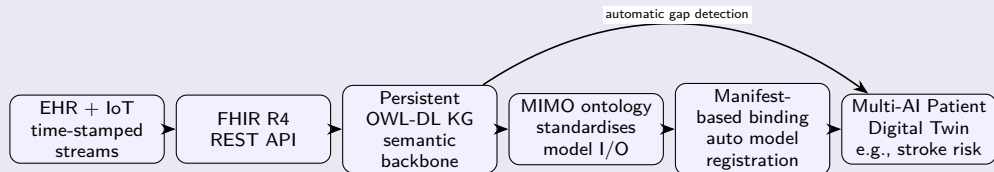
This pipeline reads CSV exports. What is the minimal change to make it stream-native — and does adding an MQTT/Kafka hop reintroduce a semantic gap the SOSA layer was meant to close?

IoT-KG-Digital-Twin: A Three-Layer Pipeline



- **Architecture (Gyrard et al., 2024):** sensor layer → semantic-annotation layer (IoT-Lite + SOSA/SSN) → KG-driven DT that models **patient mental-health trajectories**.
- **Transformation steps:** physiological signal → SOSA observation → KG node typed by SNOMED CT → **temporal SPARQL** detects burnout/depression patterns.
- **Real-time vs. batch trade-off:** continuous *emotion monitoring* is near-real-time, but **reasoning is query-driven (semi-batch)** → good for screening, insufficient for crisis intervention.

i Discussion Prompt for PhD Students



- **Architecture (Marfoggia et al., 2026):** a **FHIR R4 REST API** retrieves time-stamped EHR/IoT streams into a persistent **OWL-DL KG**; the novel **MIMO ontology** standardises every AI model's inputs/outputs for transparent, reusable composition.
- **Transformation steps:** FHIR resource → OWL-DL assertion → MIMO interface match → manifest-driven binding → multi-AI pipeline.
- **Real-time vs. batch trade-off:** real-time API *retrieval* + automatic **gap detection** when sensor coverage is insufficient, but **OWL-DL inference latency** under load is the throttle.

i Discussion Prompt for PhD Students

MIMO makes models *pluggable*. Does plug-ability erode clinical accountability — if a stroke-risk model is hot-swapped at runtime, who signs off on the prediction that triggered an intervention?

Real-Time vs. Batch: Cross-Architecture Trade-offs

Architecture	Ingestion	Materialisation	Reasoning cadence	Best fit
Pourvabab 2025 (RDF/R2RML)	real-time (distributed)	on-write	batch ML	large-scale analytics
Chatterjee 2024 (CSV→SOSA)	offline/batch	on-load	query-time	research / reproducibility
Gyrard 2024 (3-layer)	near-real-time	on-annotation	semi-batch SPARQL	screening / prevention
Marfoggia 2026 (FHIR+MIMO)	real-time API	on-retrieval	on-demand OWL-DL	clinical decision support

- **Universal pattern:** ingestion moves toward real-time; *reasoning* stays batch/periodic — the gap between them is the clinical-latency bottleneck.
- None of the four provides **streaming incremental reasoning** with bounded guarantees.

i Discussion Prompt for PhD Students

12 minutes — “A graph that thinks — but how, and is it right?”

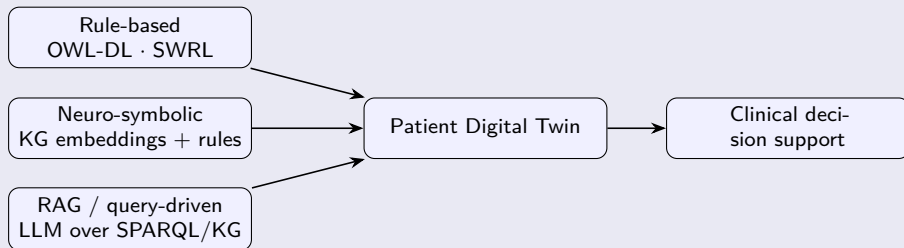
- Reasoning families in play: **rule-based** (OWL-DL / SWRL), **neuro-symbolic** / **RAG**, and **query-driven** temporal inference.
- For each: mechanism → validation methodology → **clinical applicability gap**.

i Discussion Prompt for PhD Students

“Reasoning” is doing a lot of work here. Before the slides: which mechanism would you trust to *change a drug dose* unaided — and why is your intuition probably wrong?

Reference. Krämer, B. J. (2025). *A semantic framework for patient digital twins in chronic care* [Preprint]. **arXiv**. <https://doi.org/10.48550/arxiv.2510.09134>

The Reasoning Mechanism Landscape



- **Rule-based (OWL-DL/SWRL):** transparent, auditable, but brittle & authoring-heavy — used by Marfoglia (2026), Krämer (2025), Sarani Rad (2024).
- **Neuro-symbolic / RAG:** emerging for clinical QA; trades some auditability for adaptivity — *not yet grounded* in this corpus's deployed systems.
- **Common gap across all:** scarce prospective clinical validation.

i Discussion Prompt for PhD Students

A rules engine is explainable but slow to evolve; a neuro-symbolic model adapts but hides logic. For a **regulated** PDT, which side of the trade-off does the law force you onto?

- **Reasoning mechanism (Marfoggia et al., 2026):** **OWL-DL inference** enforces semantic consistency and **auto-triggers gap detection** when sensor coverage is insufficient; the **MIMO ontology** standardises AI model I/O for dynamic, manifest-driven multi-model composition.
- **Validation:** benchmarked end-to-end on **stroke-risk prediction** with real-world, time-aware clinical streams → demonstrates KG-to-PDT scalability.
- **Clinical applicability gap:** no prospective trial; generalisation beyond stroke untested; **OWL-DL reasoning latency under concurrent load** unquantified.

i Discussion Prompt for PhD Students

MIMO guarantees *interface* compatibility, not *semantic* correctness. Two models with matching I/O can disagree ontologically — how would you detect that before a twin silently contradicts itself?

Reference. Marfoggia, A., D'Errico, C., Mellone, S., & Carbonaro, A. (2026). *A knowledge graph-driven framework for deploying AI-powered patient digital twins*. **Future Generation Computer Systems**, Article 108380. <https://doi.org/10.1016/j.future.2026.108380>

Federated SWRL Reasoning: Privacy-First Patient Twins

- **Reasoning mechanism (Krämer, 2025):** the **Patient Medical Digital Twin (PMDT)** in **OWL 2.0** with five clinical Blueprints; **SWRL rules executed at the edge** over multimodal IoT time-series; only **OWL class assertions** (never raw data) are propagated centrally.
- **Validation:** synthetic patient scenarios in **diabetes & heart failure** — checks *completeness of ontological coverage* and *soundness of temporal inference chains*.
- **Clinical applicability gap:** synthetic-only, no real deployment; SWRL coverage depends on **expert rule authoring**; federated reasoning untested at scale.

Discussion Prompt for PhD Students

Propagating only class assertions looks privacy-safe — but OWL inferences can *re-identify* a rare patient. When does “no raw data” still leak a patient?

Reference. Krämer, B. J. (2025). *A semantic framework for patient digital twins in chronic care* [Preprint]. **arXiv**. <https://doi.org/10.48550/arxiv.2510.09134>

PHKG-Driven Diabetes Twin: Reasoning on a Live Graph

- **Reasoning mechanism (Sarani Rad et al., 2024):** a **Personal Health KG (PHKG)** is the digital twin's substrate — it **auto-updates** as CGM/EHR/PROM observations arrive; **OWL reasoning infers hypoglycaemia-risk patterns** and medication-adjustment cues (HL7 FHIR for exchange).
- **Validation:** evaluated on a **real patient cohort**; superior glycaemic-outcome prediction vs. rule-based baselines.
- **Clinical applicability gap:** single disease, **retrospective cohort**, no prospective RCT; **PHKG update latency** under high-frequency CGM unaddressed.

i Discussion Prompt for PhD Students

The PHKG updates incrementally — but a stale edge can flip a hypoglycaemia inference. Design a *temporal-validity* mechanism: when does a triple “expire”?

Reference. Sarani Rad, F., Hendawi, R., Yang, X., & Li, J. (2024). *Personalized diabetes management with digital twins: A patient-centric knowledge graph approach*. **Journal of Personalized Medicine**, 14(4), Article 359. <https://doi.org/10.3390/jpm14040359>

Reasoning Cross-Check: Mental-Health Twin

- **Reasoning mechanism (Gyrard et al., 2024):** temporal SPARQL queries against the patient KG detect burnout/depression trajectories; SOSA/SSN + SNOMED CT + FHIR enable cross-domain clinical joins.
- **Validation:** proof-of-concept **use-case demonstration** linked to SDG-3 targets — *no clinical-cohort benchmark*.
- **Clinical applicability gap:** affect lacks clean SNOMED CT representation → reasoning rests on **under-specified clinical semantics**; intervention triggering unvalidated.

i Discussion Prompt for PhD Students

If the KG can't crisply represent “burnout trajectory,” is the Twin simulating a patient or merely a sensor-derived proxy? How would you measure the semantic gap?

Reference. Gyrard, A., Mohammadi, S., Gaur, M., & Kung, A. (2024). *IoT-based preventive mental health using knowledge graphs and standards for better well-being* [Preprint]. **arXiv**.
<https://doi.org/10.48550/arxiv.2406.13791>

8 minutes — “Where the graph still breaks”

- Synthesise across **all eight papers** on four axes: **scalability, temporal reasoning, privacy/federation, evaluation benchmarks**.
- Goal: convert gaps into **three PhD-defining research questions**.

i Discussion Prompt for PhD Students

Of the four axes, only **one** currently blocks clinical deployment more than the others. Which — and can you name the paper whose absence on that axis is most telling?

Reference. Bibliography compiled for “*Knowledge Graphs as Semantic Bridges between CPS/IoT and Patient Digital Twins*” (8 papers, 2023–2026), APA 7th edition — see References.

Cross-Paper Synthesis Matrix

Paper (year)	Scalability	Temporal reasoning	Privacy / federated	Evaluation
Pourvahab et al. (2025)	Yes: distributed layer	—	No	Yes: interop metrics
Gyrard et al. (2024)	Part	Yes: temporal SPARQL	No	Part: use-case demo
Marfoggia et al. (2026)	Yes: end-to-end bench	Yes: time-aware	No	Yes: stroke benchmark
Carbonaro et al. (2023)	—	Yes: identified req	Yes: privacy-by-design	— conceptual
Krämer (2025)	Part	Yes: temporal inference	Yes: class-assertion fed.	Part: synthetic only
Sarani Rad et al. (2024)	Part	Yes: incremental PHKG	No	Yes: real cohort
Chatterjee et al. (2024)	—	—	No	Yes: real vs synthetic
Hendawi & Li (2024)	Part: prototype	Yes: longitudinal	Yes: RBAC schema	Yes: real EHR

Scalability × Temporal Reasoning

- **The contradiction:** CPS streams demand *fresh* graphs; OWL-DL/SWRL reasoning is **non-incremental** → re-materialisation cost grows with graph size.
- Only **Pourvahab (2025)** and **Marfoggia (2026)** tackle scale, and neither provides **streaming incremental reasoning with bounded latency**.
- Temporal support is mostly **query-time** (SPARQL) or **scenario-based** (Krämer), not a first-class temporal model (e.g., valid-time RDF).

i Discussion Prompt for PhD Students

Design a temporal KG where a triple carries $[t_start, t_end)$ validity and reasoners update only the *delta*. What breaks first — the OWL-DL classifier, the SPARQL index, or the clinical timestamp semantics?

References. Pourvahab et al. (2025), <https://doi.org/10.1145/3756681.3757054> · Marfoggia et al. (2026), <https://doi.org/10.1016/j.future.2026.108380> · Krämer (2025), <https://doi.org/10.48550/arxiv.2510.09134>

Privacy-Preserving & Federated KGs

- **Best in class:** Krämer (2025) propagates *only OWL class assertions*; Hendawi & Li (2024) embed **RBAC in the schema**; Carbonaro (2023) elevates **privacy-by-design** to a requirement.
- **Collective gap:** no **differential privacy, secure multi-party computation, or verifiable federation** over health KGs; **inference-based re-identification** of rare patients is unanalysed.
- Consent + governance (CONNECTED's open agenda) remain **unimplemented** in any reviewed system.

i Discussion Prompt for PhD Students

Federated learning hides gradients, not *graph structure*. What does a patient's KG topology alone reveal about their diagnosis — even with zero raw values?

References. Krämer (2025), <https://doi.org/10.48550/arxiv.2510.09134> · Hendawi & Li (2024), <https://doi.org/10.1109/aimhc59811.2024.00026> · Carbonaro et al. (2023), <https://doi.org/10.3389/fdgth.2023.1322428>

Evaluation Benchmarks: The Missing Yardstick

- Each paper invents **its own** dataset/metric — interop metrics (Pourvahab), SPARQL coverage (Chatterjee), cohort AUC (Sarani Rad), scalability bench (Marfoggia), synthetic completeness (Krämer).
- There is **no shared CPS-KG-PDT benchmark** measuring semantic interoperability *and* inference latency *and* twin fidelity together.
- Synthetic-only validation (Krämer, Gyrard) cannot certify clinical soundness.

i Discussion Prompt for PhD Students

Define the **three metrics** a community benchmark must mandate so that two different KG-PDT systems become genuinely comparable. What does each metric deliberately *exclude*?

References. Pourvahab et al. (2025) · Chatterjee et al. (2024) · Sarani Rad et al. (2024) · Marfoggia et al. (2026) · Krämer (2025) · Gyrard et al. (2024) — full DOIs on the References slide.

Three Open PhD Research Questions

💡 RQ1 — Streaming Temporal Reasoning

How can we perform **incremental temporal reasoning** (valid-time RDF + delta-aware OWL/SWRL) over CPS/IoT streams with **bounded, clinically-tolerable latency** — instead of periodic batch re-materialisation?

💡 RQ2 — Verifiably-Private Federated KG Reasoning

Can we design **federated neuro-symbolic reasoning** that never exposes raw patient triples *or* re-identifying graph structure, with **provable privacy guarantees and full auditability** (GDPR Art. 22)?

💡 RQ3 — A Shared CPS-KG-PDT Benchmark

What benchmark suite (datasets, interop/latency/fidelity metrics, baselines) would let heterogeneous KG-PDT architectures be compared on equal terms — and what synthetic→real data ladder makes it clinically credible?

Key Takeaways

- **KGs are the bridge** — but the bridge's quality is set by *ontology alignment*, not sensor count.
- **Ingestion is real-time; reasoning is not** — that gap *is* the clinical-latency problem.
- **Rule-based reasoning dominates**; neuro-symbolic/RAG remain promising but ungrounded clinically.
- **Privacy and benchmarks are the twin blind spots** of the entire corpus.
- A defensible PhD lives where the matrix has a **“No” on a deployment-critical axis**.

i Discussion Prompt for PhD Students

State, in one sentence, the research question your thesis would attack — then check: does it sit on a “No” cell from the synthesis matrix? If not, is it actually PhD-sized?

Reference. Bibliography compiled for *“Knowledge Graphs as Semantic Bridges between CPS/IoT and Patient Digital Twins”* (8 papers, 2023–2026), APA 7th edition.

i Discussion Prompt for PhD Students

These eight papers share a blind spot. Name one missing voice — a method, a clinical domain, or a geography — that your thesis would add to the corpus.

- Carbonaro, A., Marfoglia, A., Nardini, F., & Mellone, S. (2023). CONNECTED: Leveraging digital twins and personal knowledge graphs in healthcare digitalization. *Frontiers in Digital Health*, 5, Article 1322428. <https://doi.org/10.3389/fdgth.2023.1322428>
- Chatterjee, A., Gerdes, M., Prinz, A., Riegler, M., & Martínez, S. (2024). Semantic representation and comparative analysis of physical activity sensor observations using MOX2-5 sensor in real and synthetic datasets: A proof-of-concept study. *Scientific Reports*, 14, Article 55183. <https://doi.org/10.1038/s41598-024-55183-6>
- Gyrard, A., Mohammadi, S., Gaur, M., & Kung, A. (2024). *IoT-based preventive mental health using knowledge graphs and standards for better well-being* [Preprint]. arXiv. <https://doi.org/10.48550/arxiv.2406.13791>
- Hendawi, R., & Li, J. (2024). Comprehensive personal health knowledge graph for effective management and utilization of personal health data. In *Proceedings of the IEEE 2nd International Conference on AI in Medicine and Health Care (AIMHC)* (pp. 1–8). IEEE. <https://doi.org/10.1109/aimhc50011.2024.00006>